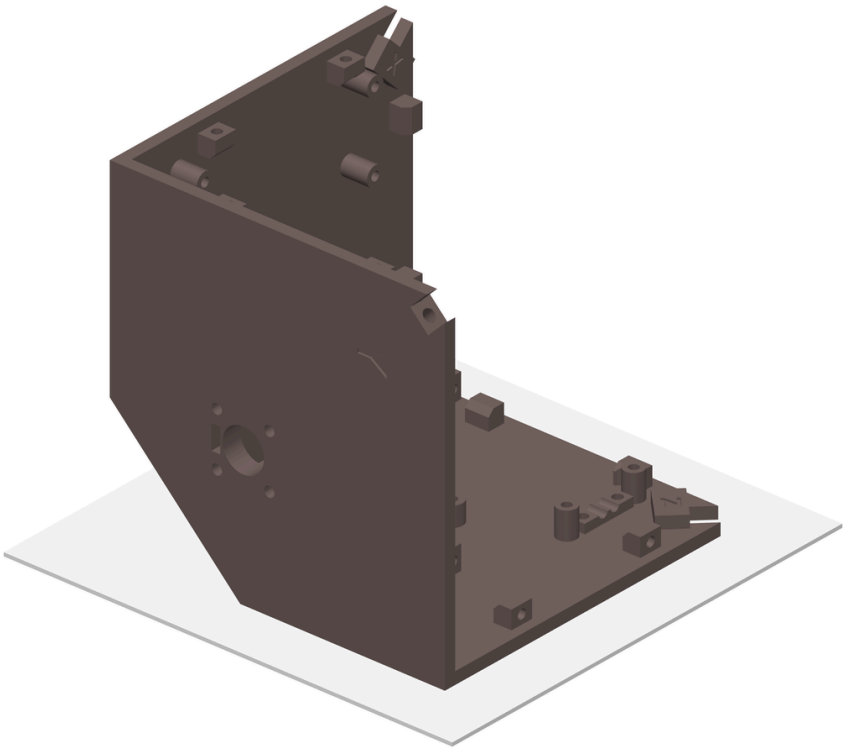
Printing

Every printed part, in the order you need it.



Designed by Athanasia Nikolova

All data come from tests and many real builds, not only from theory. v1.2 · November 2026

Teacher only



One thing per page

One action, one big picture. Short sentences, much white space.



Color is never the only clue

Every color also has a number, a letter or a name next to it. The part colors stay distinct with color-vision deficiency (Okabe-Ito palette).



Clear print

Instructions at 14 pt, everything else at least 12 pt, in Atkinson Hyperlegible. The text has a contrast of at least 4.5 : 1 to the page (WCAG 2.x, AA).



For a screen reader

The PDF is tagged (PDF/UA-1): headings, lists and reading order are marked, and every picture has a text alternative that is read aloud.



Print it right

In color, at 100 % - not 'fit to page'.



Pace

One page, then the action, then the next page. For a student who tires quickly: one lesson per session. The check at the end of each lesson is a good place to stop.

Safety first



Grown-up nearby

Work with your teacher or mentor in the room.



Hot!

The nozzle and the bed are HOT. Never touch them while the printer works or just after.



Hands out

Hands out of the printer while it moves.



Fresh air

Open a window: the warm plastic smells.



Off a cool bed

Take a part off only when the bed is cool. Push the scraper AWAY from your hand.



Dust

Carbon-fiber filament: no sanding without a mask - its dust is not for breathing.

Before you start



The filament

PETG-CF (PETG with carbon fiber). It wears a soft brass nozzle - use a hardened steel nozzle.



Keep it dry

A closed bag with a drying pack, or dry it before printing.



Two colors

Milk Tea Brown and Ancient Wood Green.



As they are

The files in CAD/print are already turned the right way: put them on the bed as they are.



Your printer's bed

At least 166 x 123 mm, and 131 mm high (the longest part and the box).

The three settings

Every part uses one of them - its page says which.


layer height


walls


infill


nozzle and bed

Standard - most parts

- Layer height 0.16 mm · 4 walls (wall loops)
- Top: 5 layers (0.8 mm) · bottom: 4 layers
- Infill 20 %, rectilinear pattern · no support

Fine - small and precise parts

- Layer height 0.08 mm · 5 walls (wall loops)
- Infill 20 %, rectilinear pattern · no support (the navel: WITH support)

Heavy - the reaction wheels

- Layer height 0.16 mm · 4 walls (wall loops)
- Infill 90 %, rectilinear pattern · no support - the wheels must be heavy

All three

- Speeds: first layer 50 mm/s · inner wall 220 mm/s · outer wall 170 mm/s
- PETG-CF: nozzle 255 °C · bed 75 °C · fan 40 %
- Dry the filament 8 hours at 65 °C, and print straight from the dryer.

How to read the pages



The picture

The part as it lies on the bed.



Under the title

How many you print, and which filament (its color dot and its name).



Settings

Layer height, walls, infill, support - set them in your slicer.



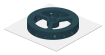
Grams

From the 3D model (or weighed, where it says so). Good for planning the filament.

All the parts

In the order the build needs them.

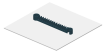
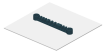
1 · The box and the wheels

-  Box ×1 — 172 g
-  Reaction wheel ×3 — 282 g
-  Magnet carrier ×3 — 2 g

2 · Battery holder and IMU

-  Battery tube ×1 — 22 g
-  Battery basin ×1 — 30 g

3 · Wiring

-  clamp_power ×1 — 0 g
-  clamp_z ×1 — 2 g
-  clamp_y ×1 — 3 g
-  clamp_x_1 ×1 — 1 g
-  clamp_x_2 ×1 — 1 g
-  Clamp spacer ×8 — 1 g

4 · The cage

-  Box standoff ×1 — 6 g
-  Hub standoff ×1 — 3 g

-  Cage corner ×8 — 21 g
- Cage beam ×9 — 83 g
- Tie beam ×3 — 27 g

5 · Covers

-  Cover ×3 — 58 g

6 · The sphere

-  Navel ×8 — 66 g
-  Edge arc ×12 — 90 g
-  Diagonal arc, lower ×6 — 77 g
-  Diagonal arc, upper ×6 — 77 g

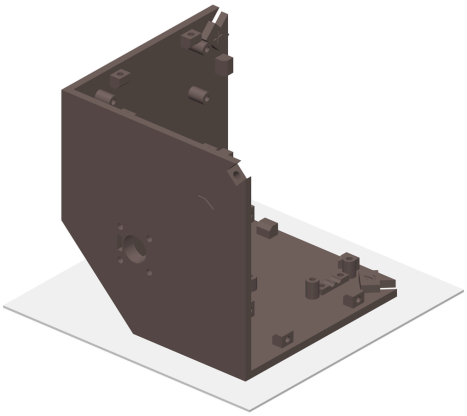
Filament, all together

- **Ancient Wood Green** — about 467 g
- **Milk Tea Brown** — about 556 g

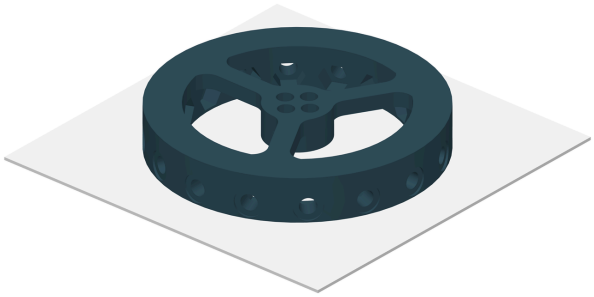
Plus about 10 % for the first layers, supports and a failed print.

1 · The box and the wheels

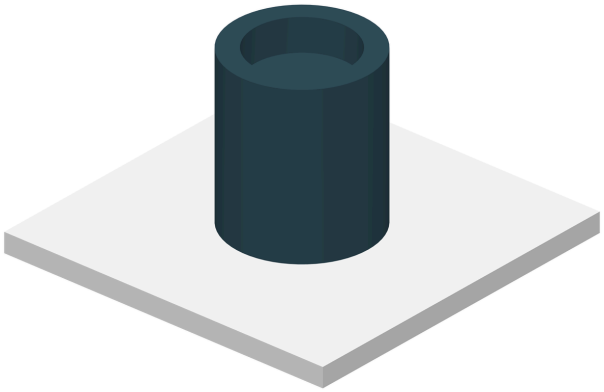
Chapter 1 of the Assembly guide.



Box ×1



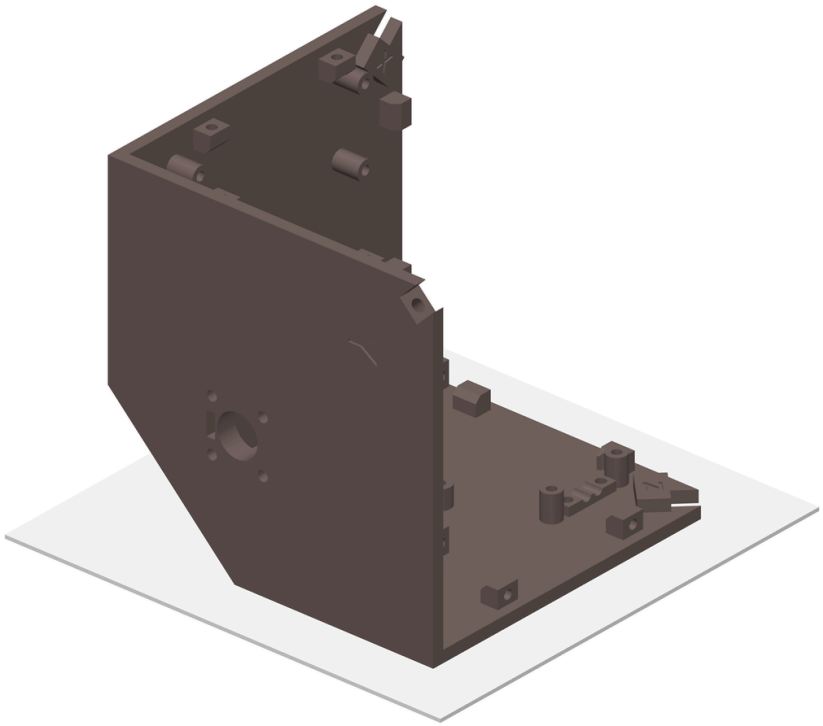
Reaction wheel ×3



Magnet carrier ×3

Box

×1 Milk Tea Brown · PETG-CF

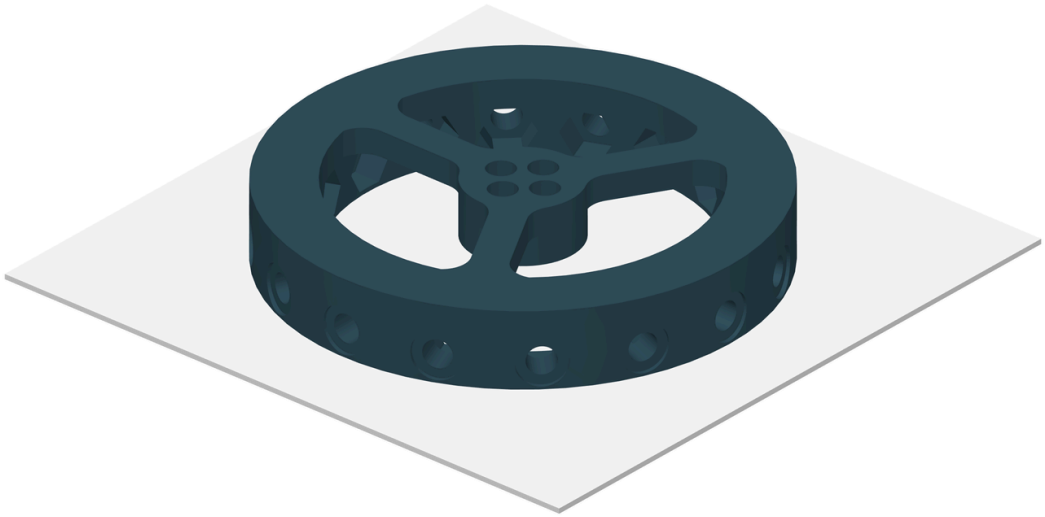


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 172 g each.
- The biggest print: it needs the whole height of the printer.
- It was printed and weighed: 172 g with these settings.

File: CAD/print/box.step

Reaction wheel

×3 Ancient Wood Green · PETG-CF

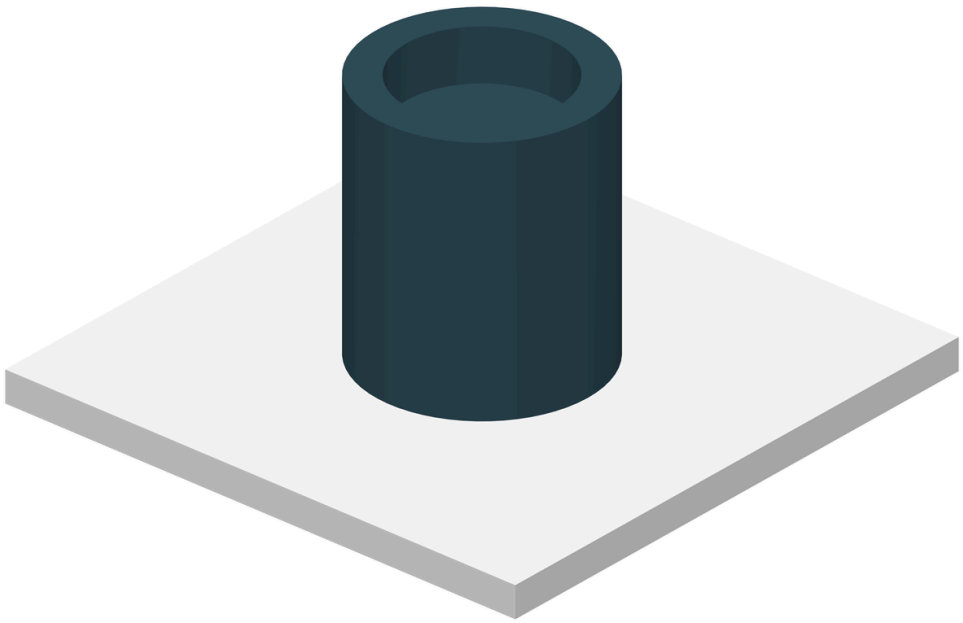


- Layer 0.16 mm · 4 walls · 90% infill · no support - the heavy settings
- About 94 g each · 282 g for all 3.
- Flat on the bed, as in the picture.
- Heavy settings (90 % infill): its weight is what balances the cube.

File: CAD/print/reaction_wheel.step

Magnet carrier

×3 Ancient Wood Green · PETG-CF



- Layer 0.08 mm · 5 walls · 20% infill · no support - the fine settings
- About 1 g each · 2 g for all 3.
- Fine settings: it holds the magnet exactly in the middle.

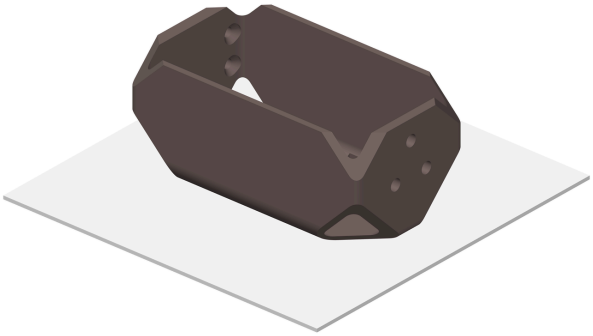
File: CAD/print/magnet_carrier.step

2 · Battery holder and IMU

Chapter 2.



Battery tube ×1



Battery basin ×1

Battery tube

×1 Ancient Wood Green · PETG-CF

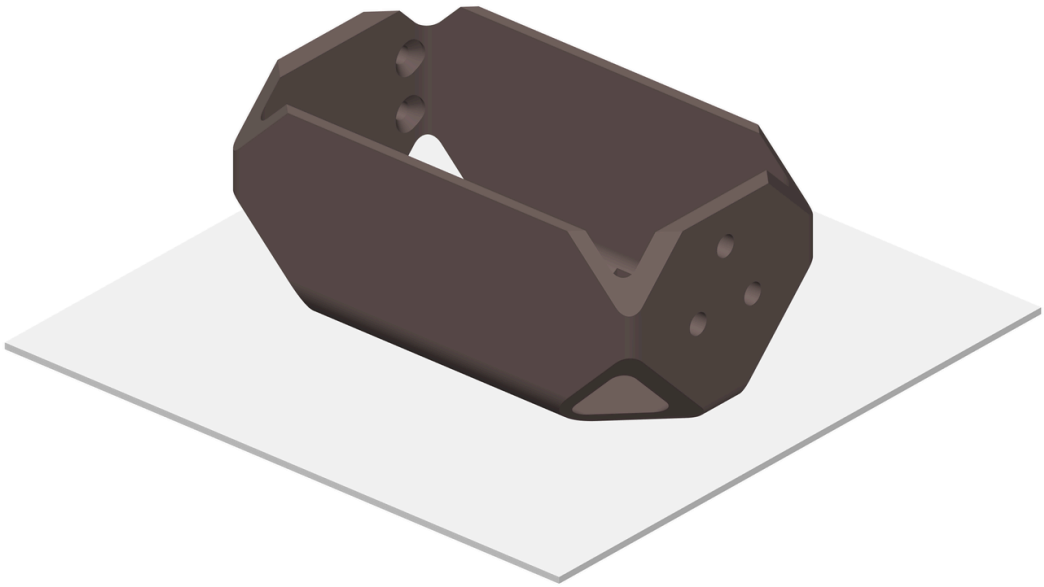


- Layer 0.16 mm · 4 walls · 100% infill · no support
- About 22 g each.
- SOLID (100 % infill): the IMU sits in its table - it must not bend or shake.
- The foot on the bed.

File: CAD/print/battery_tube.step

Battery basin

×1 Milk Tea Brown · PETG-CF

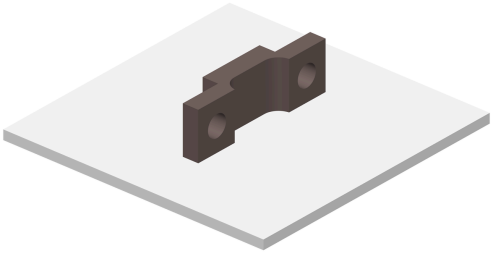


- Layer 0.16 mm · 4 walls · 100% infill · no support
- About 30 g each.
- Print it ON ITS BACK, as in the picture.

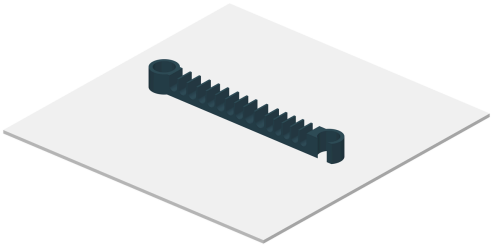
File: CAD/print/battery_basin.step

3 · Wiring

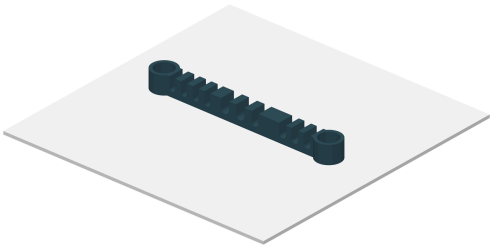
Chapter 3 - clamp_power, the 4 cable clamps and their spacers.



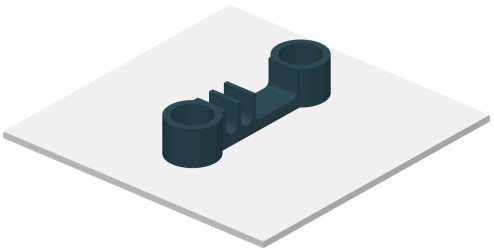
clamp_power ×1



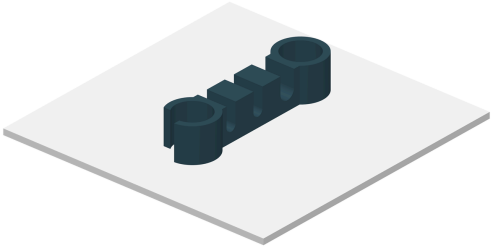
clamp_z ×1



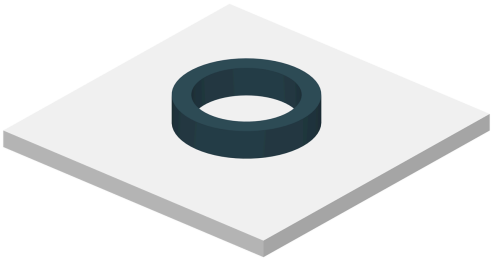
clamp_y ×1



clamp_x_1 ×1



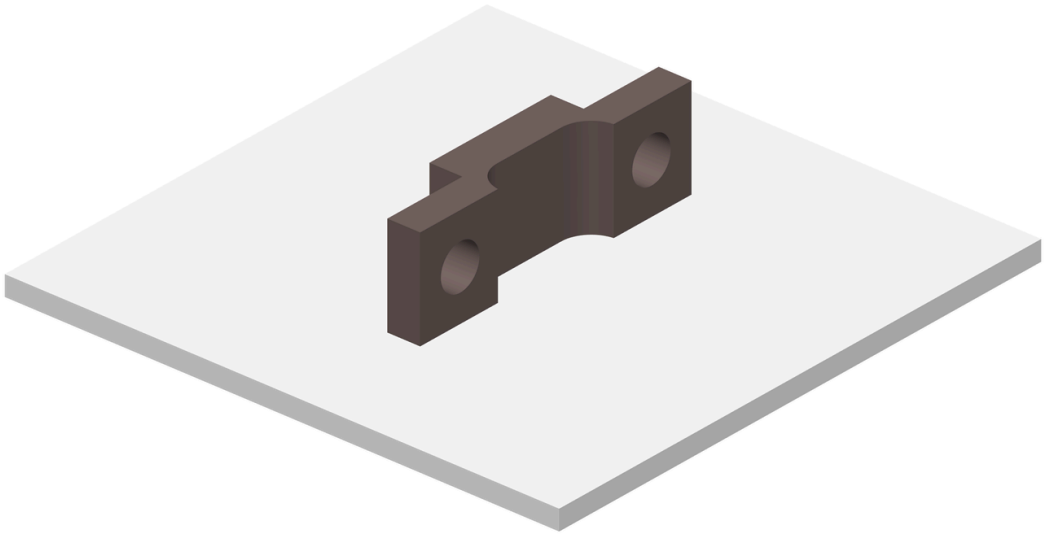
clamp_x_2 ×1



Clamp spacer ×8

clamp_power

×1 Milk Tea Brown · PETG-CF

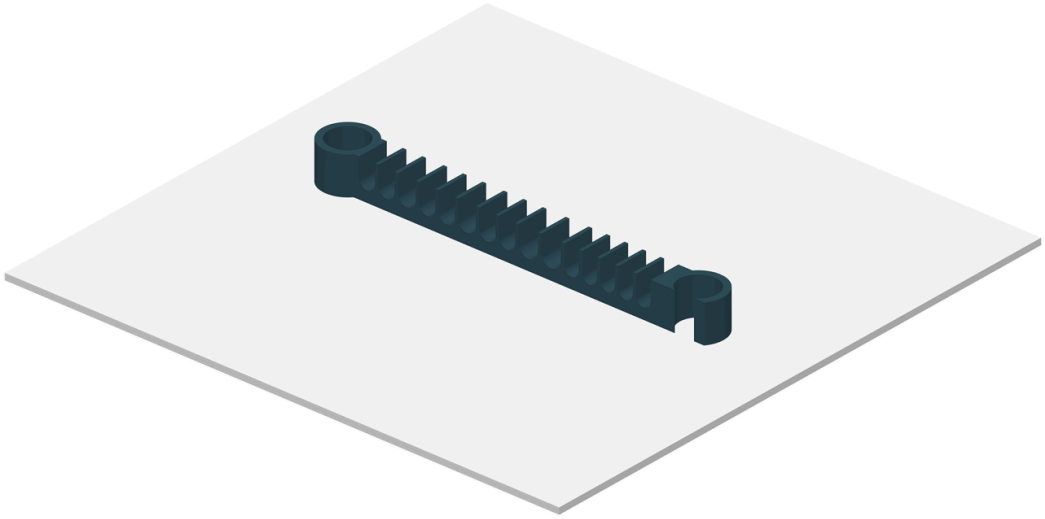


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 0 g each.

File: CAD/print/clamp_power.step

clamp_z

×1 Ancient Wood Green · PETG-CF

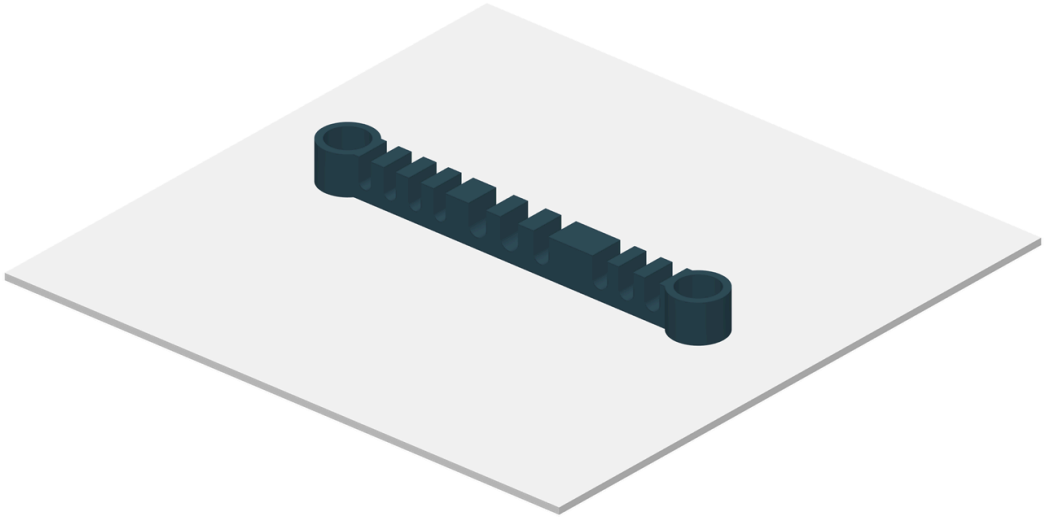


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 2 g each.
- Upside down, as in the picture: the flat top on the bed, the openings up.
- No support: the openings are open channels.

File: CAD/print/clamp_z.step

clamp_y

×1 Ancient Wood Green · PETG-CF

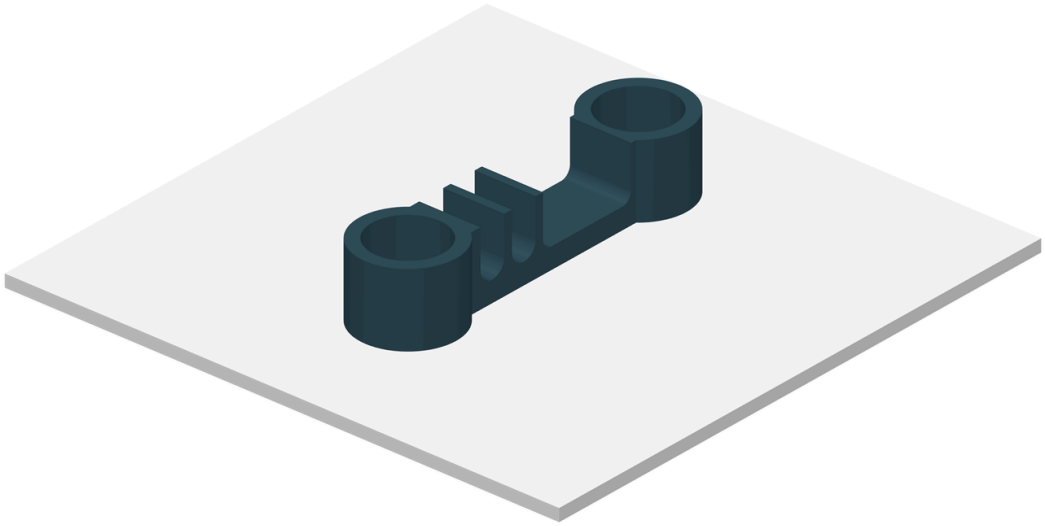


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 3 g each.
- Upside down, as in the picture: the flat top on the bed, the openings up.
- No support: the openings are open channels.

File: CAD/print/clamp_y.step

clamp_x_1

×1 Ancient Wood Green · PETG-CF

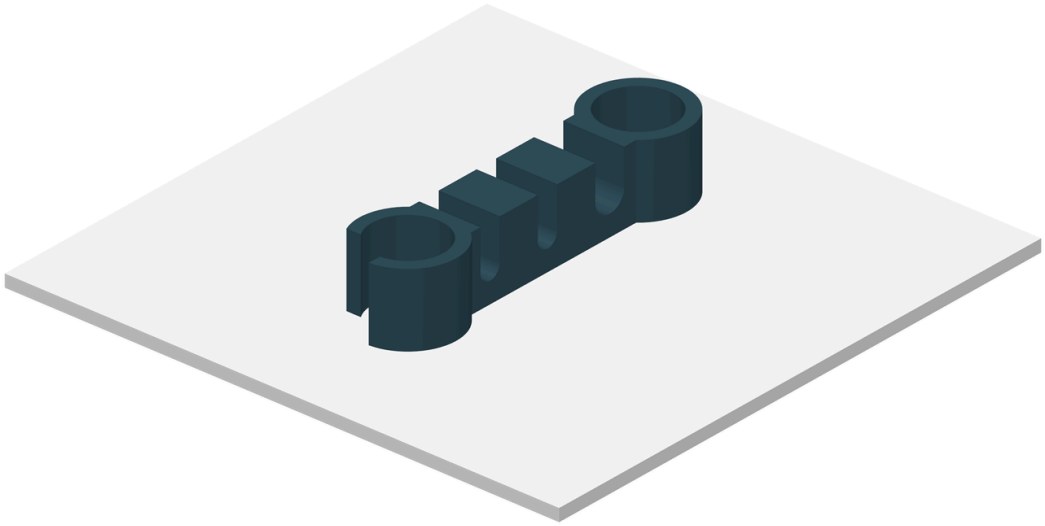


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 1 g each.
- Upside down, as in the picture: the flat top on the bed, the openings up.
- No support: the openings are open channels.

File: CAD/print/clamp_x_1.step

clamp_x_2

×1 Ancient Wood Green · PETG-CF

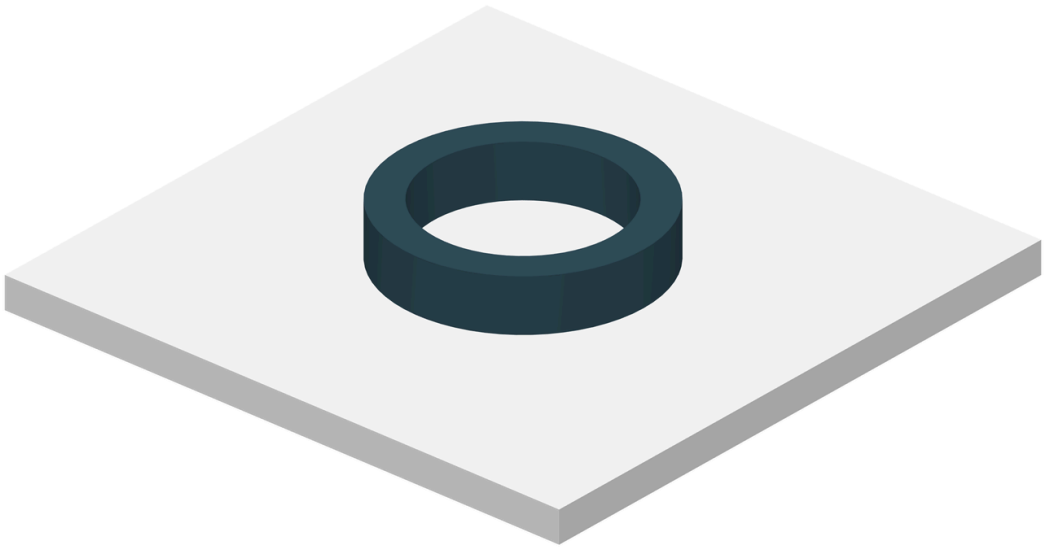


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 1 g each.
- Upside down, as in the picture: the flat top on the bed, the openings up.
- No support: the openings are open channels.

File: CAD/print/clamp_x_2.step

Clamp spacer

×8 Ancient Wood Green · PETG-CF

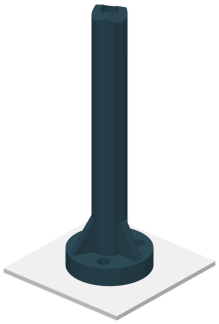


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 0 g each · 1 g for all 8.
- 8 of them: one on each clamp ring, under the board.
- Small: print them together with a bigger part.

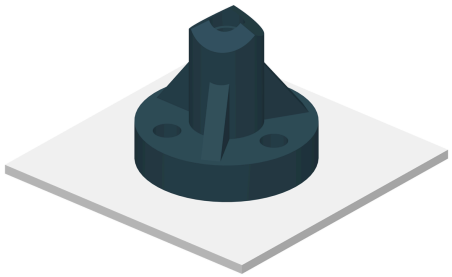
File: CAD/print/spacer_OD9.5_ID7_H2.step

4 · The cage

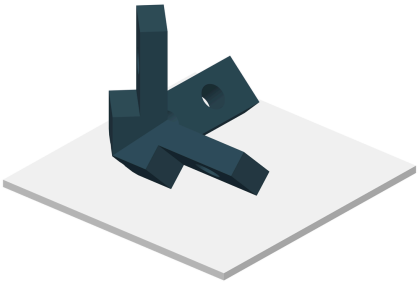
Chapter 4.



Box standoff ×1



Hub standoff ×1



Cage corner ×8



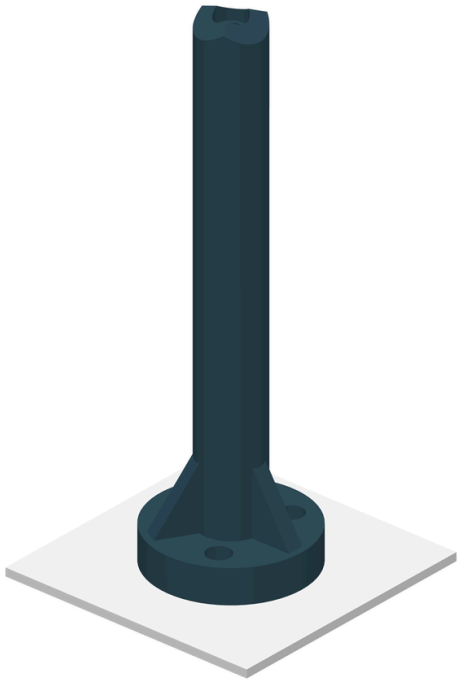
Cage beam ×9



Tie beam ×3

Box standoff

×1 Ancient Wood Green · PETG-CF

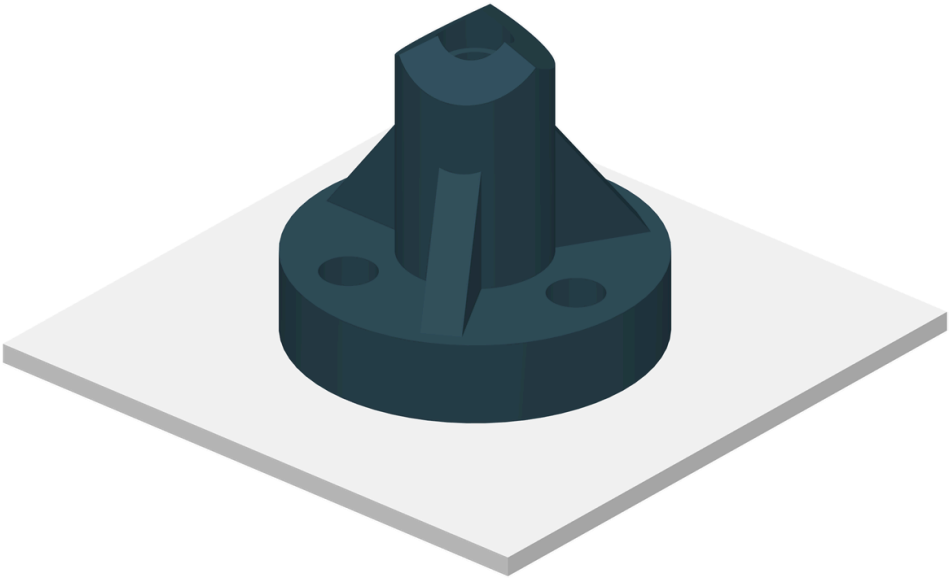


- Layer 0.08 mm · 5 walls · 20% infill · no support - the fine settings
- About 6 g each.

File: CAD/print/box_standoff.step

Hub standoff

×1 Ancient Wood Green · PETG-CF

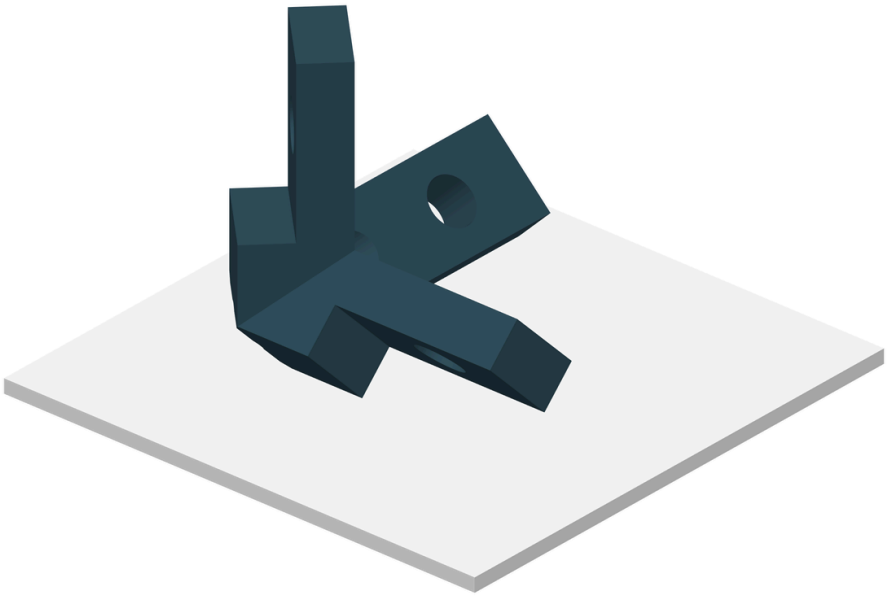


- Layer 0.08 mm · 5 walls · 20% infill · no support - the fine settings
- About 3 g each.

File: CAD/print/battery_top_standoff.step

Cage corner

×8 Ancient Wood Green · PETG-CF



- Layer 0.08 mm · 5 walls · 20% infill · no support - the fine settings
- About 3 g each · 21 g for all 8.

File: CAD/print/corner.step

Cage beam

×9 Milk Tea Brown · PETG-CF

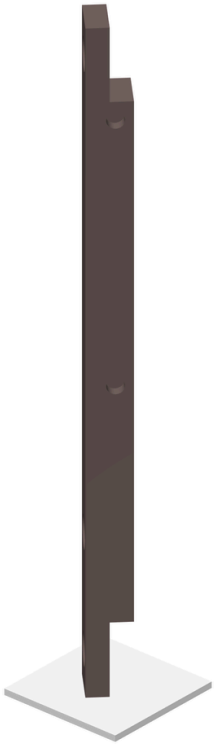


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 9 g each · 83 g for all 9.
- 9 plain beams. The other 3 beams are tie beams - the next part.

File: CAD/print/beam.step

Tie beam

×3 Milk Tea Brown · PETG-CF

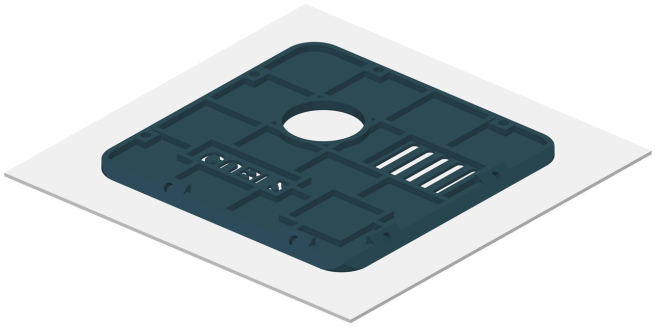


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 9 g each · 27 g for all 3.
- 3 tie beams: a hole for the box screw and 2 cover magnet pockets.

File: CAD/print/beam_tie.step

5 · Covers

Chapter 5.



Cover ×3

Cover

×3 Ancient Wood Green · PETG-CF

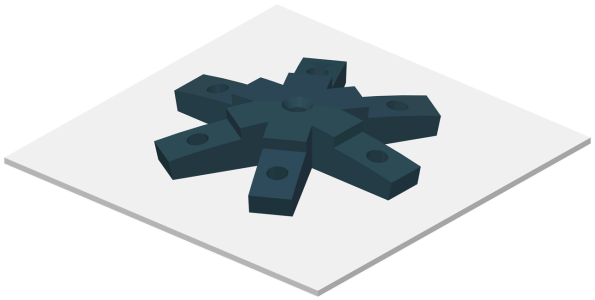


- Layer 0.08 mm · 5 walls · 20% infill · no support - the fine settings
- About 19 g each · 58 g for all 3.
- Fine settings: flat and exact, the magnets sit in it.
- Let it cool on the bed to room temperature before you take it off - then it stays flat.

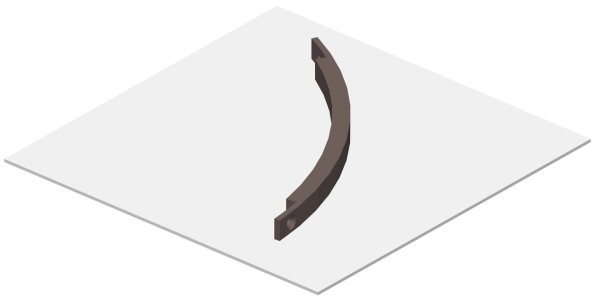
File: CAD/print/cover.step

6 · The sphere

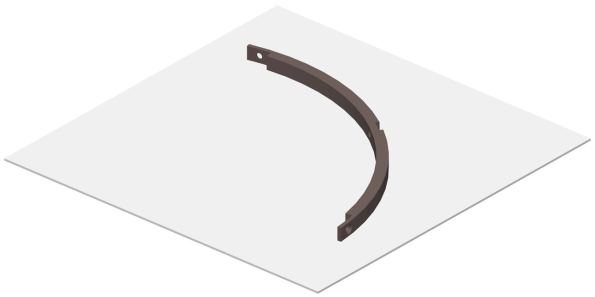
Chapter 6.



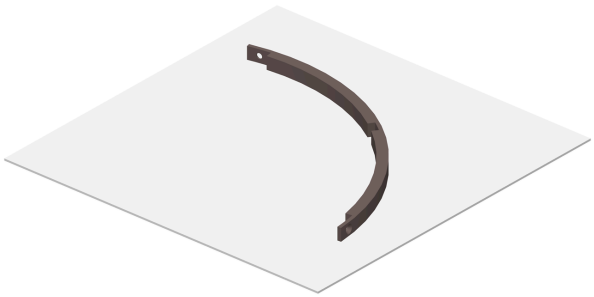
Navel ×8



Edge arc ×12



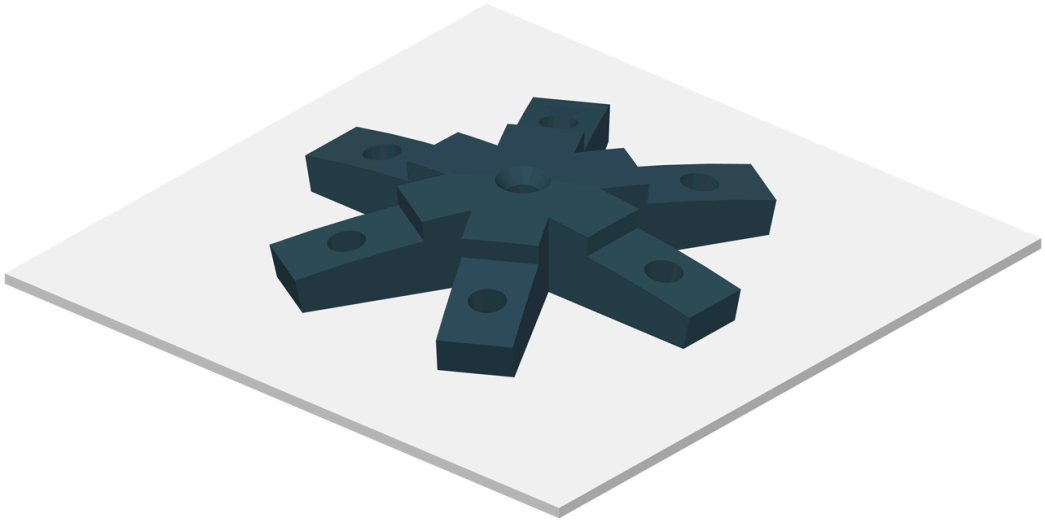
Diagonal arc, lower ×6



Diagonal arc, upper ×6

Navel

×8 Ancient Wood Green · PETG-CF

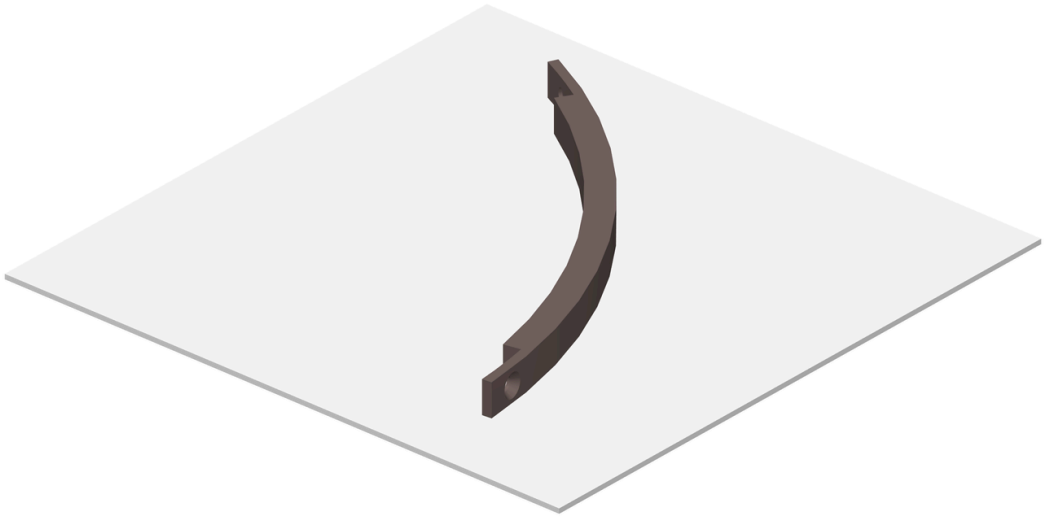


- Layer 0.08 mm · 5 walls · 20% infill · WITH support - the fine settings
- About 8 g each · 66 g for all 8.
- WITH support - the only part that needs it.

File: CAD/print/navel.step

Edge arc

×12 Milk Tea Brown · PETG-CF

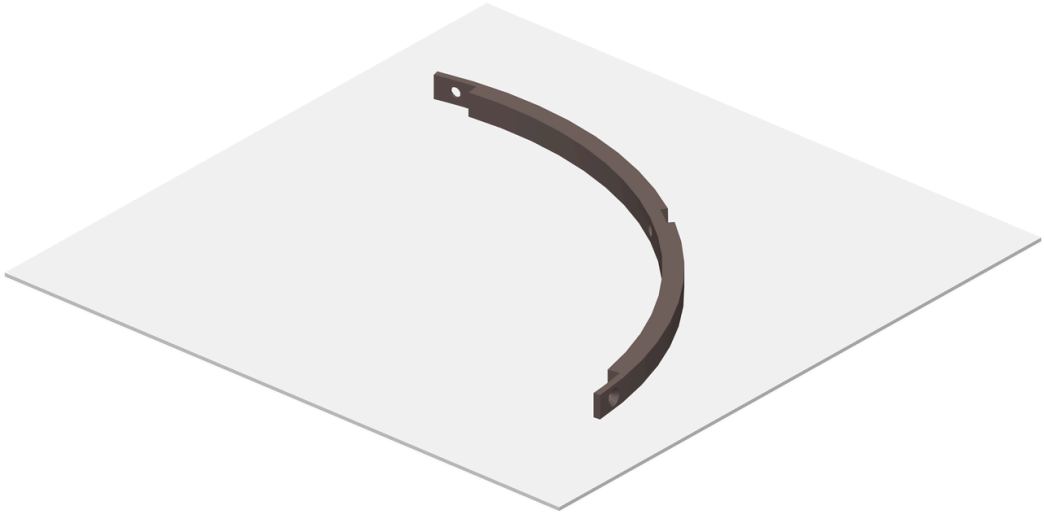


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 8 g each · 90 g for all 12.

File: CAD/print/sphere_arc_edge.step

Diagonal arc, lower

×6 Milk Tea Brown · PETG-CF

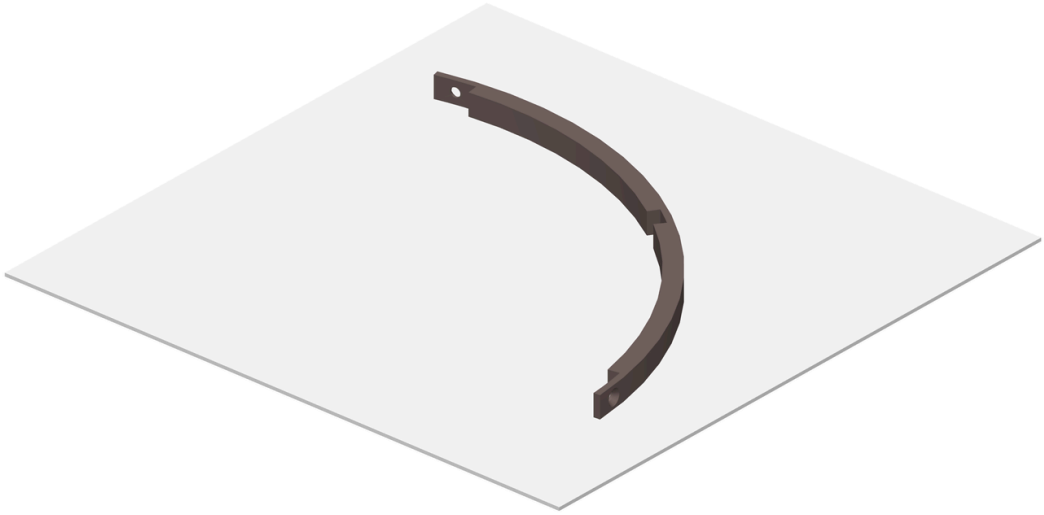


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 13 g each · 77 g for all 6.
- 6 LOWER diagonal arcs - they have the insert in the middle.

File: CAD/print/sphere_arc_diag_under.step

Diagonal arc, upper

×6 Milk Tea Brown · PETG-CF

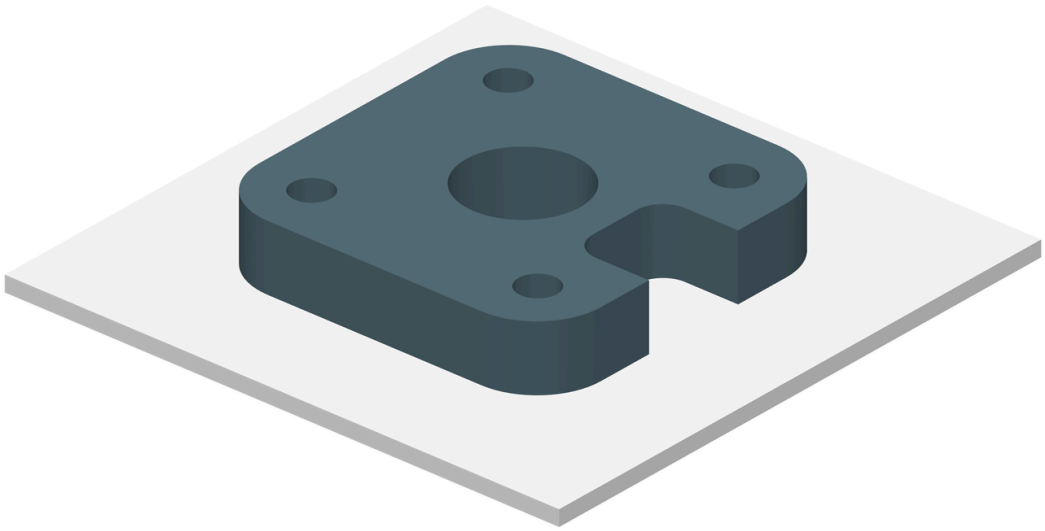


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 13 g each · 77 g for all 6.
- 6 UPPER diagonal arcs.

File: CAD/print/sphere_arc_diag_over.step

Magnet-handler (tool)

×1 any color · PETG-CF or PLA

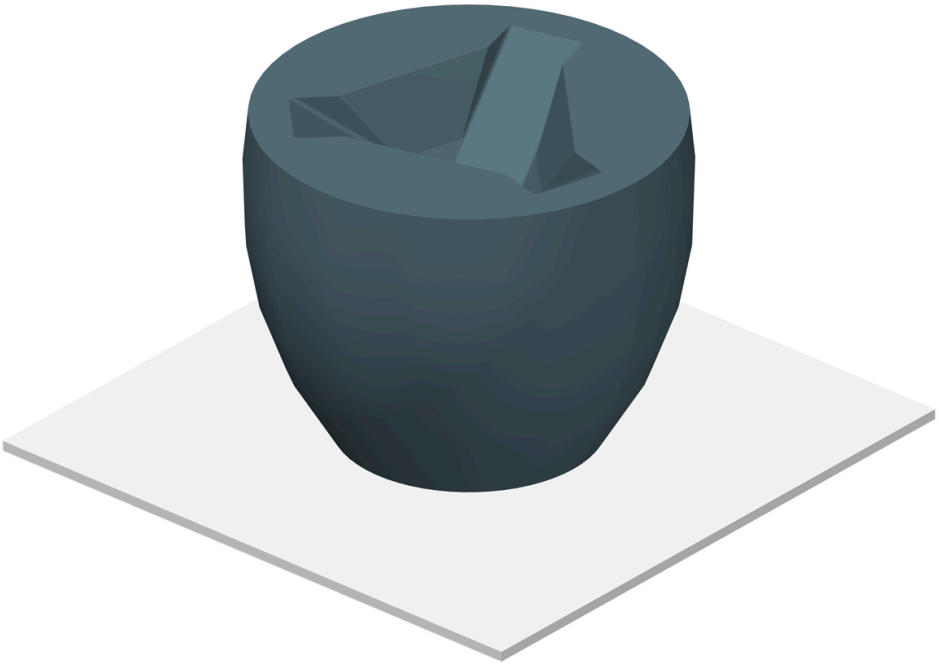


- Layer 0.08 mm · 5 walls · 20% infill · no support - the fine settings
- Magnets guide: centers the magnet carrier.
- A tool: it is used, then put away. Any color.

File: CAD/print/magnet-handler.step

Display base, top (not part of the cube)

×1 any color · PETG-CF or PLA

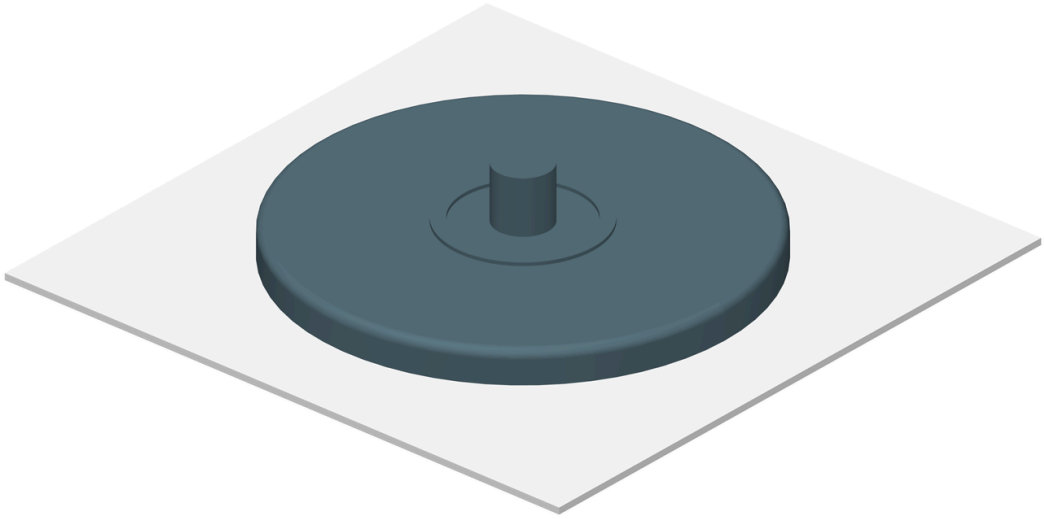


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- the cube stands on corner 0 in its cup.
- Display base, top: hollow inside, closed. Flat side on the bed.

File: CAD/print/base_top.step

Display base, bottom (not part of the cube)

×1 any color · PETG-CF or PLA

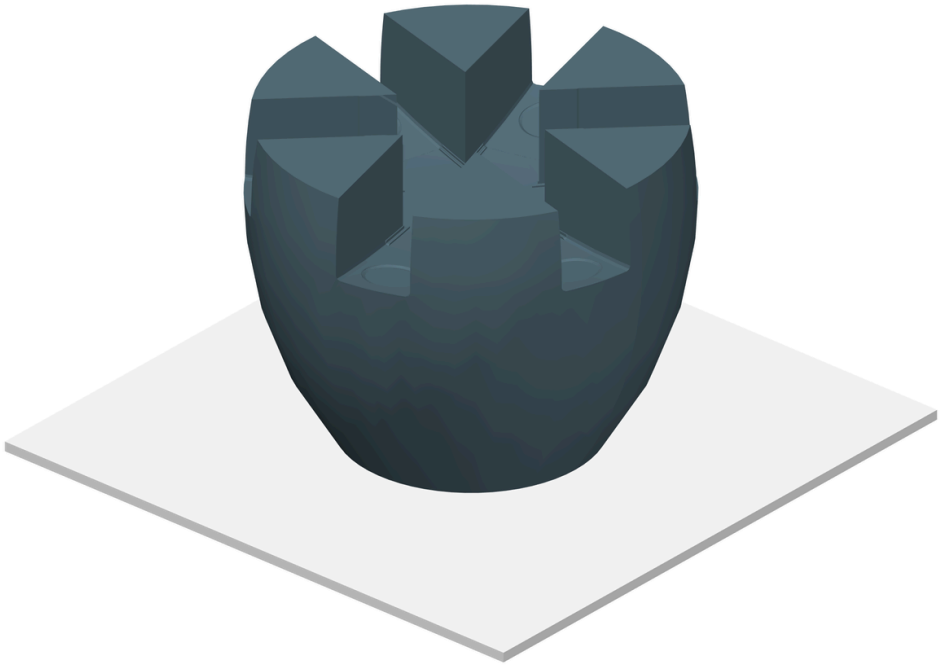


- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- the disc; the top turns on its pin.
- Display base, bottom: the disc. Flat side on the bed, the pin up.

File: CAD/print/base_bottom.step

Display base, top for the cube WITH the sphere (not part of the cube)

×1 any color · PETG-CF or PLA



- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- the navel and its 6 arcs sit in it; the same bottom.
- Display base, top for the cube with the sphere: hollow inside, closed.
- Flat side on the bed. Print base_top OR this one - the bottom is the same.

File: CAD/print/base_top_sphere.step

After printing



Off a cool bed

Wait until the bed is cool, then take the part off.



Supports off

Navels only - with pliers, away from your hand.



Every hole

No strings, no blobs. A drill bit by hand cleans a hole.



Tick each part

Write the color and the count on a list - tick each part.



Next

The Inserts guide - the brass inserts go into the printed parts.